Problem 6.2

Jack has deposited \$1,000 into a savings account. He wants to withdraw it when it has grown to \$2,000. If the interest rate is 4% annual interest compounded annually, how long will he have to wait?

We use the compound interest formula:

$$A = P(1+r)^t,$$

where

$$A = 2000, \qquad P = 1000, \qquad r = 0.04.$$

Substituting into the formula:

$$2000 = 1000(1.04)^t.$$

Dividing both sides by 1000,

$$2 = (1.04)^t.$$

Taking natural logarithms on both sides,

$$\ln(2) = t\ln(1.04).$$

Solving for t,

$$t = \frac{\ln(2)}{\ln(1.04)}.$$

Therefore,

$$t \approx 17.67299 \text{ years}.$$

So, Jack must wait approximately

$$\boxed{17.67 \text{ years}}.$$